Circuit Tracing: Revealing Computational Graphs in Language Models

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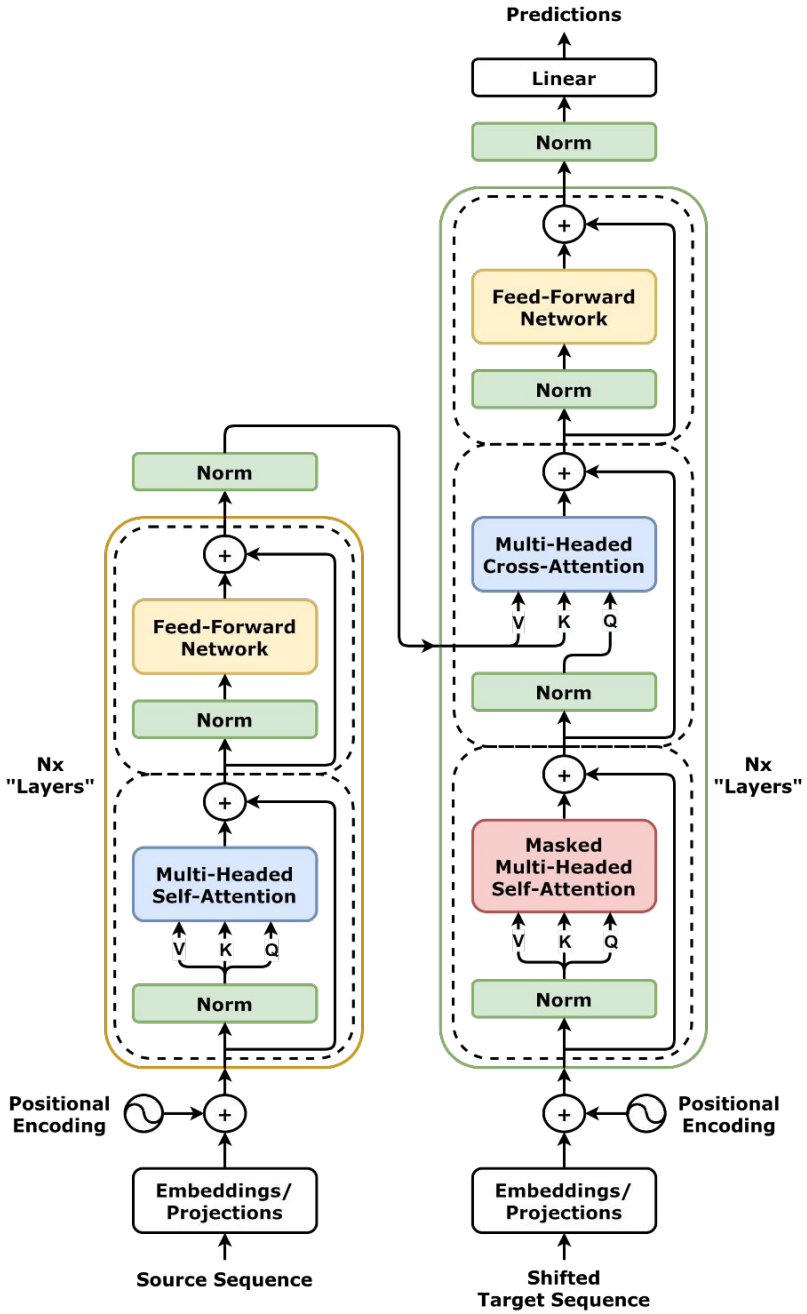
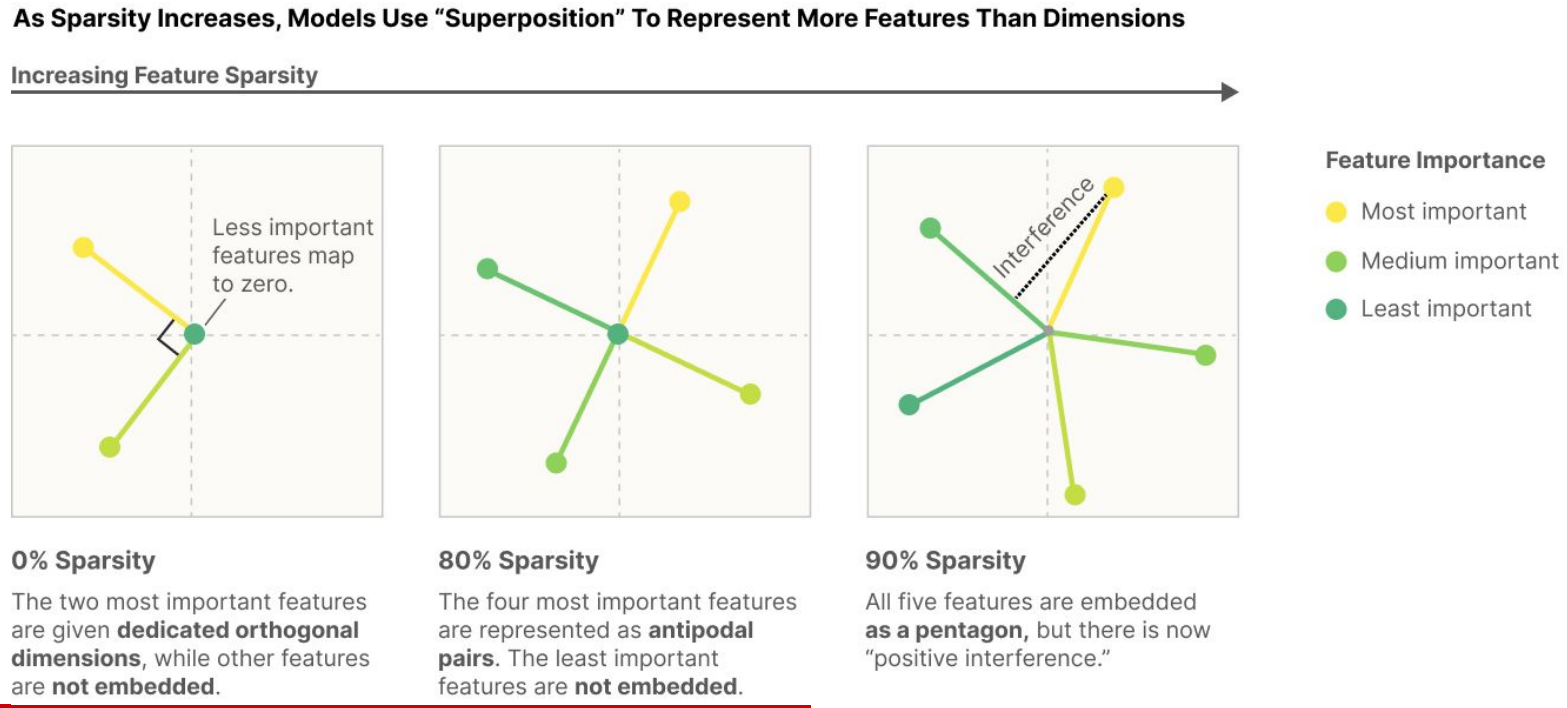
Recap: Transformer



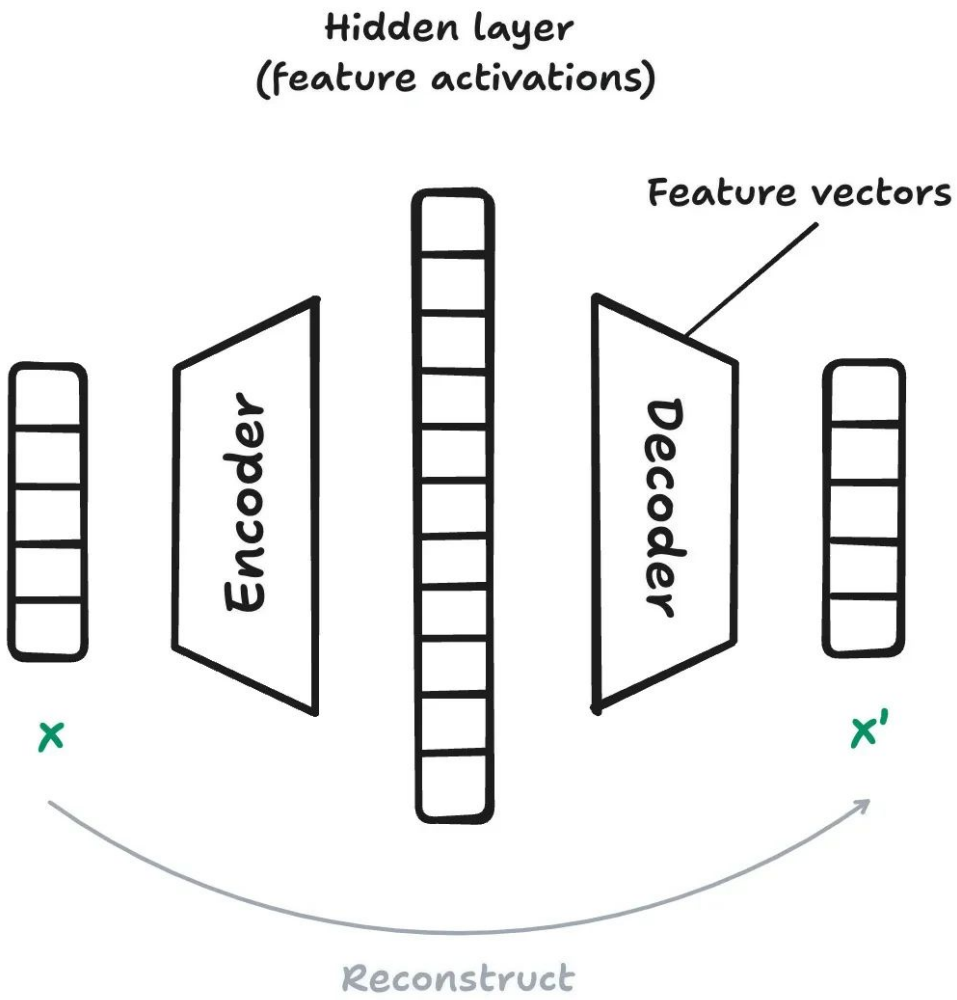
Why Sparsity?

Superposition hypothesis: neural networks represent more features than they have neurons by allowing features to be non-orthogonal. (e.g., polysemanticity of a word)



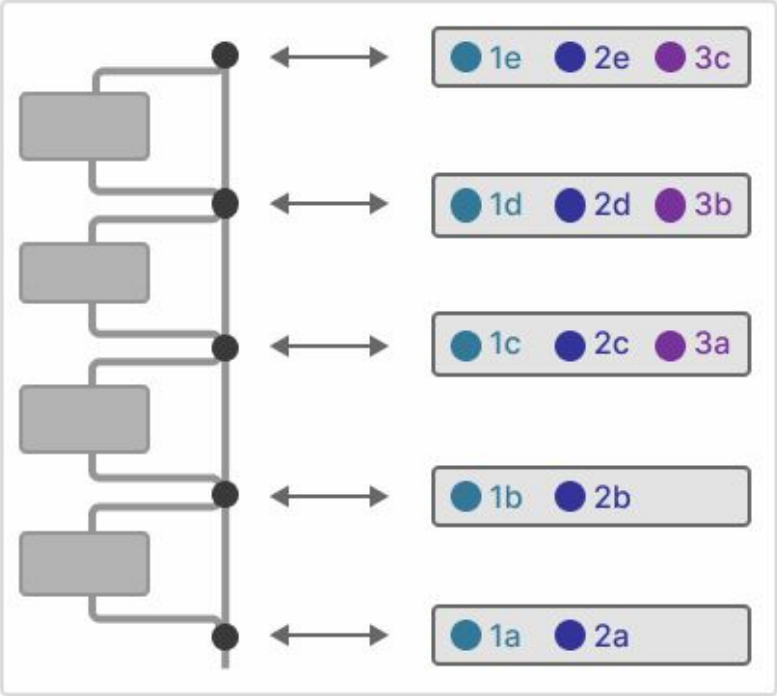
Sparse Autoencoder and Transcoder

$$\bar{\mathbf{x}} = \mathbf{x} - \mathbf{b}_d$$
$$\mathbf{f} = \mathrm{ReLU}(W_e \bar{\mathbf{x}} + \mathbf{b}_e)$$
$$\hat{\mathbf{x}} = W_d \mathbf{f} + \mathbf{b}_d$$
$$\mathcal{L} = \frac{1}{|X|} \sum_{\mathbf{x} \in X} ||\mathbf{x} - \hat{\mathbf{x}}||_2^2 + \lambda ||\mathbf{f}||_1$$

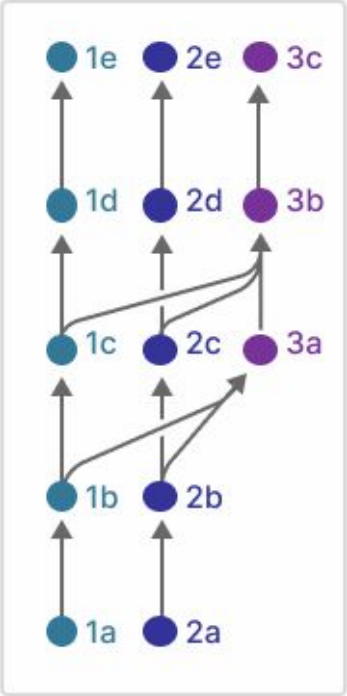


Sparse Autoencoder and Transcoder

Model with Per-Layer SAEs

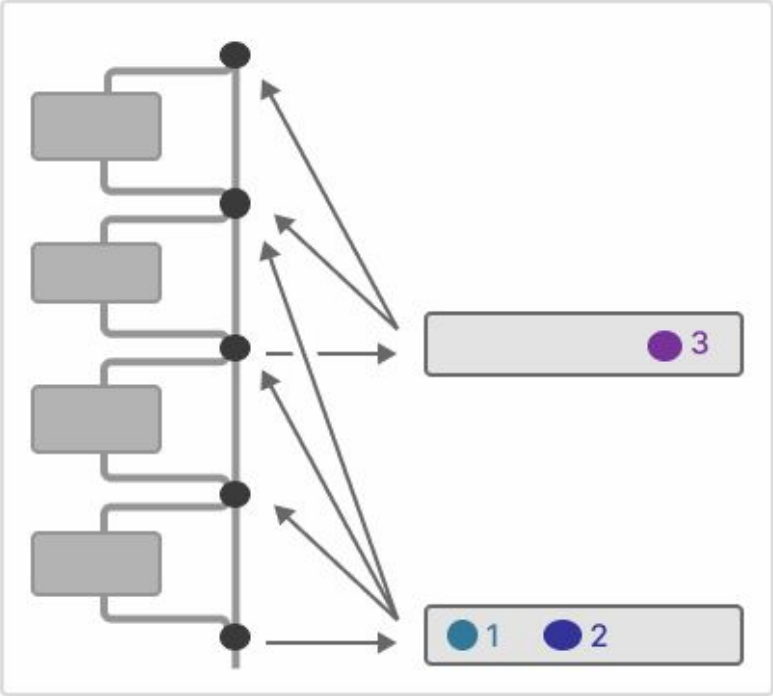


Example Circuit

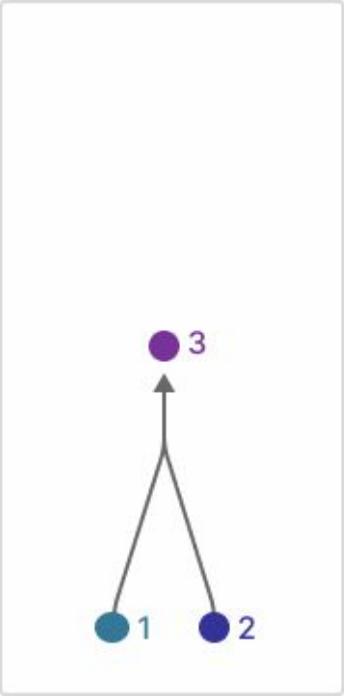


Per-layer SAEs trained on the residual stream often learn "duplicate features" because features persist in the residual stream for many layers once created. This creates a complex circuit as well.

Model with Crosscoders



Example Circuit

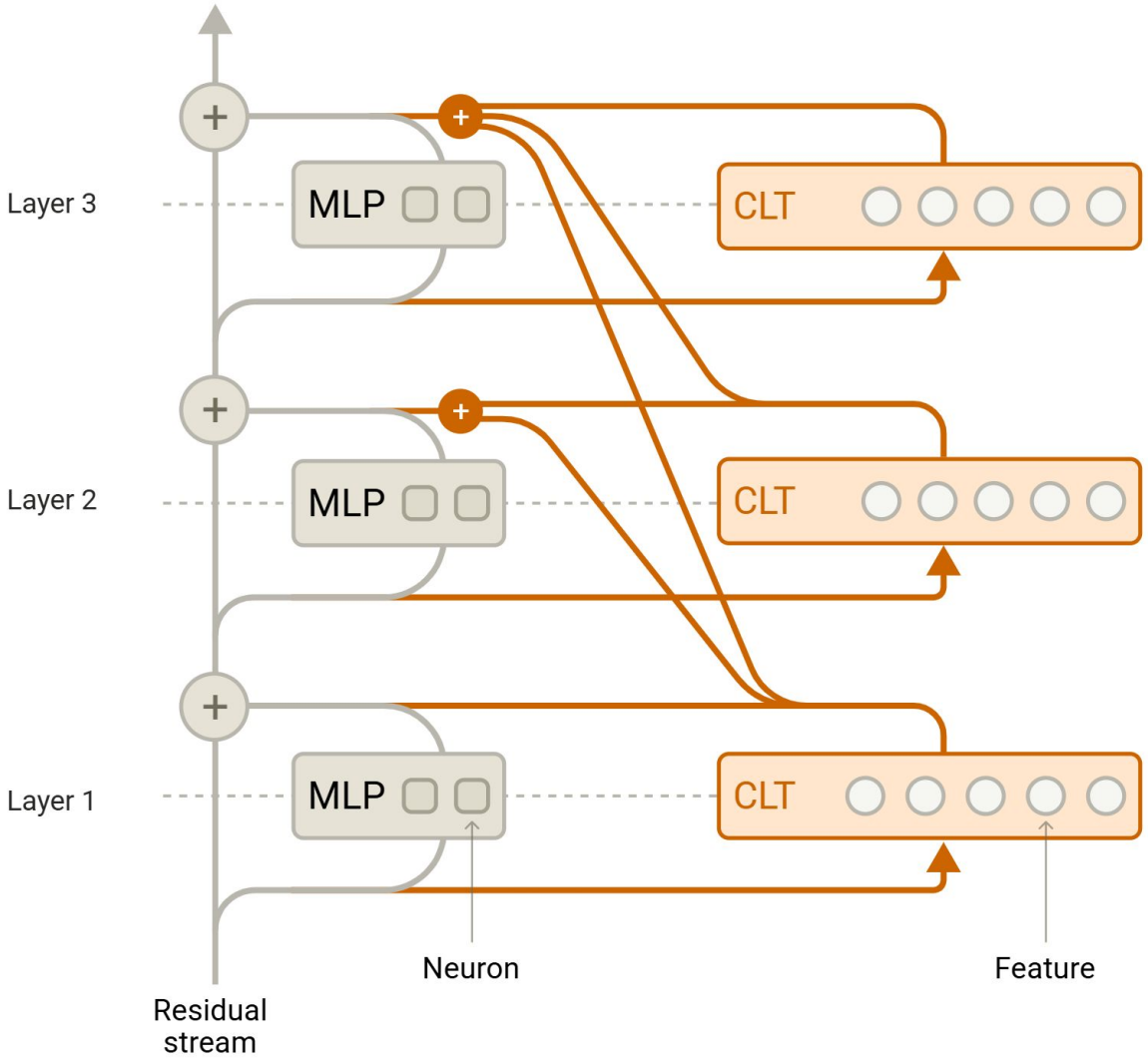


Crosscoders unify features across layers. This can potentially greatly simplify circuits.

Replacement Model

Cross-Layer Transcoder

Features read from one layer and write to all following ones



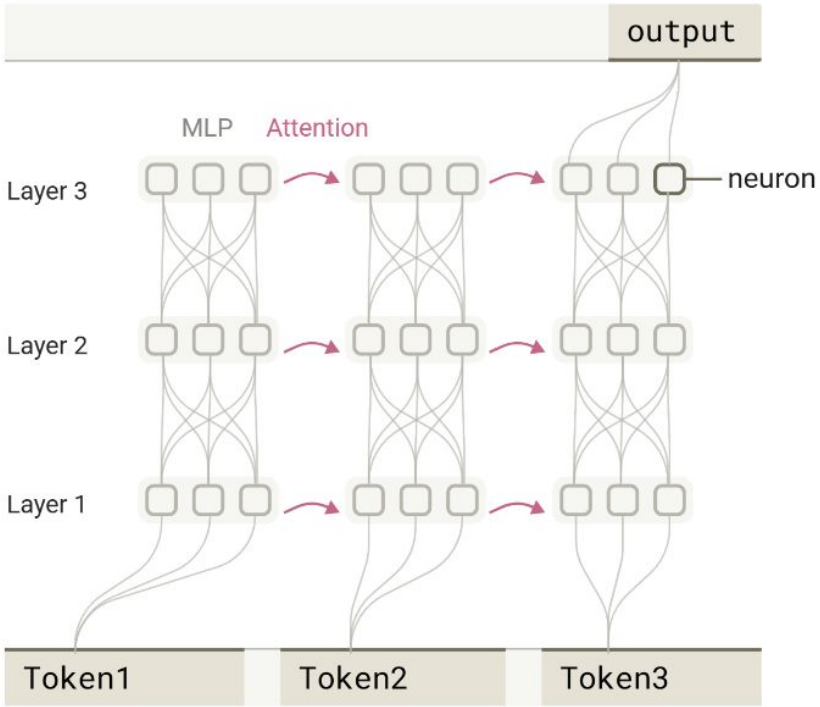
Replacement Model

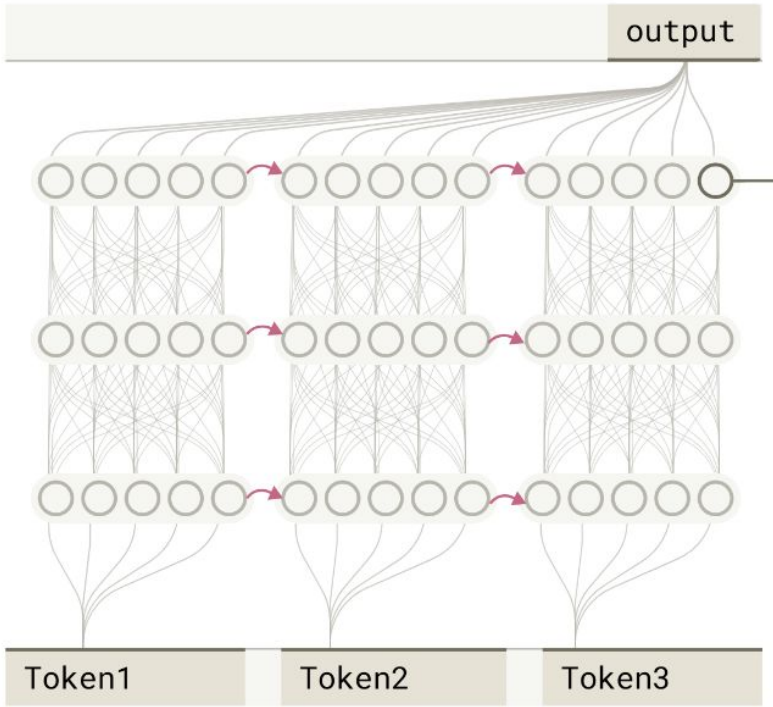
Original Transformer Model

The underlying model that we study is a transformer-based large language model.

Replacement Model

We replace the neurons of the original model with *features*. There are typically more features than neurons. Features are sparsely active and often represent interpretable concepts.





Feature

To understand what a feature represents, we use a *feature visualization*, which shows dataset examples for which the feature is most strongly active. In this example, the feature fires strongly when the model is about to say a state capital.

Figure 2: The replacement model is obtained by replacing the original model's neurons with the cross-layer transcoder's sparsely-active features.

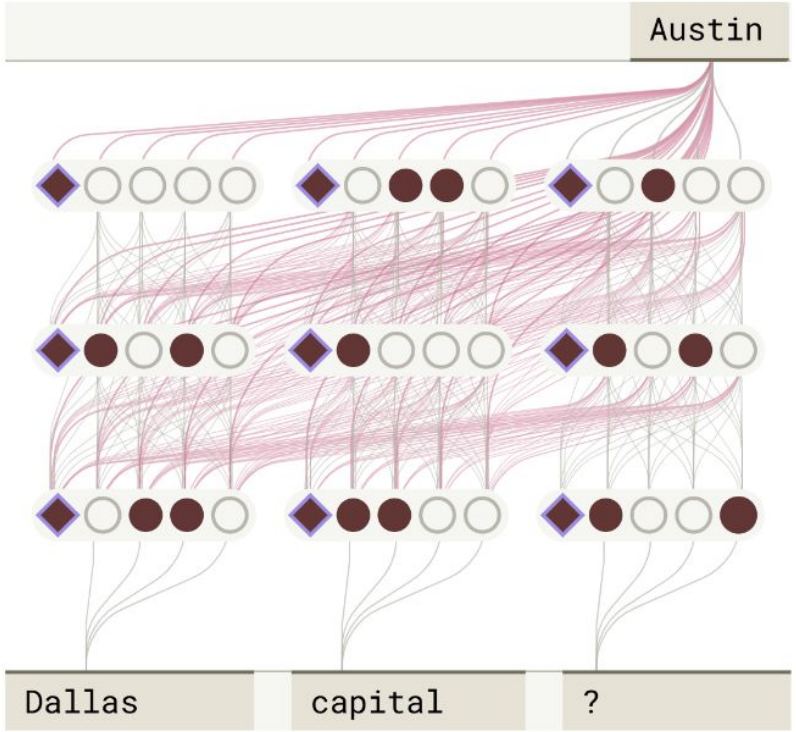
Local Replacement Model for a Fixed Prompt

- Replace MLP by CLT
- Freeze attention and normalize nonlinearities
- Adjust error at each (token position, layer) pair

Local Replacement Model for a Fixed Prompt

Local Replacement Model

The local replacement model is specific to a prompt of interest. We add an error adjustment term, and freeze attention patterns to be what they were in the original model on the given prompt. It produces the exact same output as the original model, but replaces as much computation as possible with features.



Reconstruction Error

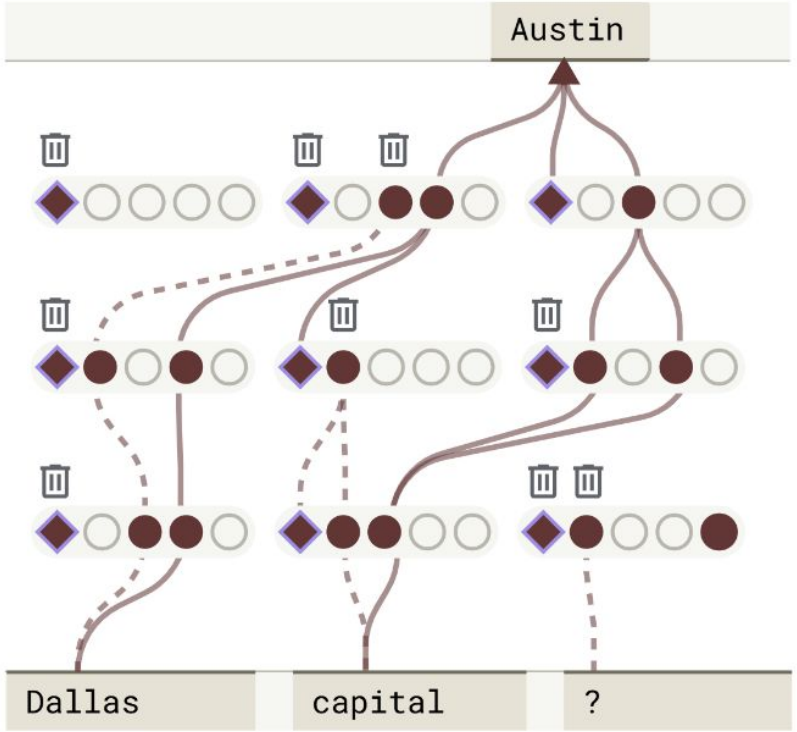
Error nodes represent the difference between the original MLP output and the replacement model's reconstruction

Attention-mediated weights

Attention patterns are frozen to their value in the original model, allowing us to define weights between features in different token positions

Attribution Graph

We trace from input to output through active features, pruning paths that don't influence the output.



Attribution Graph

https://transformer-circuits.pub/2025/attribution-graphs/methods.html

Example: Fact: the capital of the state containing Dallas is _____

https://transformer-circuits.pub/2025/attribution-graphs/biology.html

Limitation Expensive!!!!

- **Missing Attention Circuits** – We don't explain *how* attention patterns are computed by QK-circuits, and can sometimes "miss the interesting part" of the computation as a result.
- **Reconstruction Errors & Dark Matter** – We only explain a portion of model computation, and much remains hidden. When the critical computation is missing, attribution graphs won't reveal much.
- **The Role of Inactive Features & Inhibitory Circuits** – Often the fact that certain features *weren't active* is just as interesting as the fact that others are. In particular, there are many interesting circuits which involve features inhibiting other features.
- **Graph Complexity** – The resulting attribution graphs can be very complex and hard to understand.
- **Features at the Wrong Level of Abstraction** – Issues like feature splitting and absorption mean that features often aren't at the level of abstraction which would make it easiest to understand the circuit.
- **Difficulty of Understanding Global Circuits** – Ideally, we want to understand models in a global manner, rather than attributions on a single example. However, global circuits are quite challenging.
- **Mechanistic Faithfulness** – When we replace MLP computation with transcoders, how confident are we that they're using the same mechanisms as the original MLP, rather than something that's just highly correlated with the MLP's outputs?